

2425-MA140 Engineering Calculus

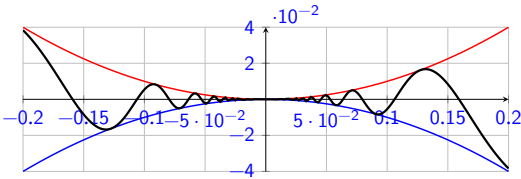
Week 2, Lectures 3

The Squeeze Theorem

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This version of the slides are by Niall Madden, but are adapted from original notes by Dr Kirsten Pfeiffer.

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- 2 Recall... Limits
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 - $\sin(\theta)/\theta$
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For more, see Section 7.8.1 (Limit of a function of a real number) in *Modern Engineering Mathematics*:
https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9780273742517

Assignment 1

- **Assignment 1** has started! You can access it on Canvas... 2425-MA140... Assignments.
- Deadline: 5pm, Friday 4 Oct 2024. (Note: that's just the deadline, you can actually start before then!)
- The **Tutorial Sheet** is available at https://universityofgalway.instructure.com/files/2040359/download?download_frd=1

Tutorials started **this** week. The schedule is on the Canvas "Course Information" page: <https://universityofgalway.instructure.com/courses/35693/pages/2425-ma140-information>

Support is also available at **SUMS**...

Recall... Limits

Yesterday, we learned that

$$\lim_{x \rightarrow a} f(x) = L,$$

means that we can make $f(x)$ as close to L as we like, by taking x as close to a as needed.

Crucially, we are usually interested in finding the limit of $f(x)$ as $x \rightarrow a$, when a is not in the domain of f .

A typical example of this is when we evaluate a rational function:

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$$

where **both** $p(a) = 0$ and $q(a) = 0$. **Idea:** Since we care about the value of p and q **near** $x = a$, but not actually at $x = a$, it is safe to factor out and $(x - a)$ term from both.

Limits of rational functions

Example

Evaluate Consider

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{p(x)}{q(x)}$$
$$p(x) = x^2 + x - 2 \quad q(x) = x^2 - x$$

Check $p(1) = 1^2 + 1 - 2 = 0$ $q(1) = 1 - 1 = 0$.

So we factorise $p(x) = x^2 + x - 2 = (x+2)(x-1)$.

$q(x) = x^2 - x = x(x-1)$.

So $\frac{x^2 + x - 2}{x^2 - x} = \frac{(x+2)(x-1)}{x(x-1)} = \frac{x+2}{x}$ f.. $x \neq 1$

$\lim_{x \rightarrow 1} \frac{p(x)}{q(x)} = \lim_{x \rightarrow 1} \frac{x+2}{x} = \frac{3}{1} = 3$ ✓

Limits of rational functions

In that last example, we found that

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x+2}{x}$$

But these are different functions:

$x=1$ is not in the domain of $\frac{x^2 + x - 2}{x^2 - x}$ but is in the domain of $\frac{x+2}{x}$.

Limits of rational functions

Evaluate the limit

$$\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x - 2} \right) = \lim_{x \rightarrow 2} \frac{p(x)}{q(x)}$$

where $p(x) = \frac{1}{2} - \frac{1}{x}$

$q(x) = x - 2$

Note $p(2) = \frac{1}{2} - \frac{1}{2} = 0$

$q(2) = 2 - 2 = 0$

Note, however, $\frac{1}{2} - \frac{1}{x} = \frac{x}{2x} - \frac{2}{2x} = \frac{x-2}{2x}$.

Then $\left(\frac{1}{2} - \frac{1}{x}\right) \cdot \frac{1}{x-2} = \left(\frac{x-2}{2x}\right) \left(\frac{1}{x-2}\right)$

If $x \neq 2$, this is $\frac{1}{2x}$.

So $\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x-2} \right) = \lim_{x \rightarrow 2} \frac{1}{2x} = \frac{1}{4}$.

More limits

Very often, we'll evaluate limits of the form:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where f and g are not polynomials. Some of the same ideas still apply.

Example

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2}$$

Idea: Simplify by completing the square in the numerator.

More limits

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2} & \quad (\text{Note if } f(x) = \frac{\sqrt{1+x^2} - 1}{x^2}, \text{ then } f(0) = \frac{0}{0} \text{ Not defined!}) \\ \text{However} \quad \frac{\sqrt{1+x^2} - 1}{x^2} &= \frac{(\sqrt{1+x^2} - 1)(\sqrt{1+x^2} + 1)}{x^2 (\sqrt{1+x^2} + 1)} \\ &= \frac{(\sqrt{1+x^2})^2 - 1}{x^2 (\sqrt{1+x^2} + 1)} = \frac{1+x^2 - 1}{x^2 (\sqrt{1+x^2} + 1)} \\ &= \frac{x^2}{x^2 (\sqrt{1+x^2} + 1)} = \frac{1}{\sqrt{1+x^2} + 1} \quad \text{for } x \neq 0. \\ \text{So } \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2} &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x^2} + 1} = \frac{1}{2} \end{aligned}$$

More limits

Exercises

Exercise 2.4

(From 2023/2024 MA140 exam, Q1(a)) Evaluate the limit

$$\lim_{x \rightarrow 4} \frac{x-4}{(\sqrt{x}-2)(x+9)}$$

The Squeeze Theorem

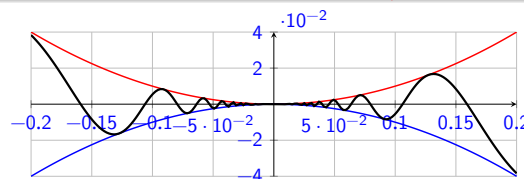
There are various approaches to evaluating limits. One significant one is...

The Squeeze Theorem (a.k.a. Sandwich Theorem)

Suppose that for functions f , g and h in a given interval I :

$$g(x) \leq f(x) \leq h(x) \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$.
So, if $f(x)$ is between g & h and g, h have some limit, so too does f .



The Squeeze Theorem

Example

Suppose $f(x)$ is a function such that

$$1 - \frac{x^2}{4} \leq f(x) \leq 1 + \frac{x^2}{2}, \quad \forall x \neq 0$$

Find $\lim_{x \rightarrow 0} f(x)$. $g(x)$ and $h(x)$

Since $\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} 1 - \frac{x^2}{4} = 1$.

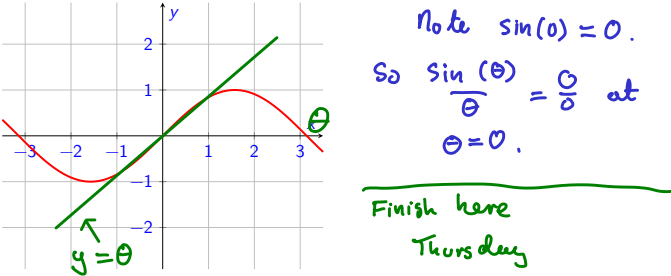
Similarly $\lim_{x \rightarrow 0} h(x) = 1$ So

$$\lim_{x \rightarrow 0} f(x) = 1.$$

We use the Squeeze Theorem to explain an important limit:

lim_{θ→0} sin θ / θ = 1

Before we show this is true, let's convince ourselves:



Before we use the Squeeze Theorem, we need a few facts about trigonometric functions.

- ▶ In this module, we only ever use radians (never, ever degrees).
- ▶ Given the triangle drawn below, $\sin \theta = \frac{b}{h}$, $\cos \theta = \frac{a}{h}$,
 $\tan \theta = \frac{b}{a} = \frac{\sin \theta}{\cos \theta}$
- ▶ Area of a sector of a circle is $\frac{1}{2}r^2\theta$ where r is the radius of the circle, and θ is the angle subtended by the sector.

Various other facts are summarised in the State Examination Commission's Tables:

Triantánacht

Trigonometry

$\tan A = \frac{\sin A}{\cos A}$

$\cot A = \frac{\cos A}{\sin A}$

$\sec A = \frac{1}{\cos A}$

$\operatorname{cosec} A = \frac{1}{\sin A}$

$\cos^2 A + \sin^2 A = 1$

$\sec^2 A = 1 + \tan^2 A$

$\cos(-A) = \cos A$

$\sin(-A) = -\sin A$

$\tan(-A) = -\tan A$

Nóta: Bíonn $\tan A$ agus $\sec A$ gan sainiú nuair $\cos A = 0$.
Bíonn $\cot A$ agus $\operatorname{cosec} A$ gan sainiú nuair $\sin A = 0$.

Note: $\tan A$ and $\sec A$ are not defined when $\cos A = 0$.
 $\cot A$ and $\operatorname{cosec} A$ are not defined when $\sin A = 0$.

A (céimeanna)	0°	90°	180°	270°	30°	45°	60°	A (degrees)
A (raidiain)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	A (radians)
$\cos A$	1	0	-1	0	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\cos A$
$\sin A$	0	1	0	-1	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$\sin A$
$\tan A$	0	-	0	-	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\tan A$

$1 \text{ rad.} \approx 57.296^\circ$

$1^\circ \approx 0.01745 \text{ rad.}$

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The Squeeze Theorem

sin(θ)/θ

The Squeeze Theorem

sin(θ)/θ

The Squeeze Theorem

sin(θ)/θ

Foirmlí uillíneacha comhsuite	Compound angle formulae
$\cos(A + B) = \cos A \cos B - \sin A \sin B$	$\cos(A - B) = \cos A \cos B + \sin A \sin B$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$	$\sin(A - B) = \sin A \cos B - \cos A \sin B$
$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$	$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

Foirmlí uillíneacha dúbailte	Double angle formulae
$\cos 2A = \cos^2 A - \sin^2 A$	$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
$\sin 2A = 2 \sin A \cos A$	$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$	$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$
$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$	

Iolraigh a thiontú ina suimeanna agus ina ndifríochtaí	Products to sums and differences
$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$	
$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$	
$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$	
$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$	

Suimeanna agus difríochtaí a thiontú ina n-iolraigh	Sums and differences to products
$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$	
$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$	
$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$	
$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$	

Here are plots of $\sin \theta$ (red) and $\cos \theta$ (blue).

MA140 — The Squeeze Theorem

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MA140 — The Squeeze Theorem

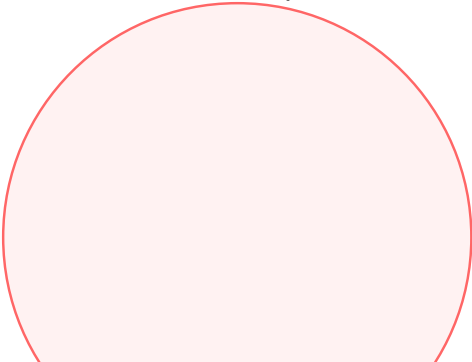
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MA140 — The Squeeze Theorem

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$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

Now let's reason more carefully:



Example

Evaluate $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}$

Example

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$